

- 5 The human gas exchange system is responsible for the efficient uptake of oxygen into the blood and for the excretion of carbon dioxide.

The different tissues of the human gas exchange system are each adapted to their specific function.

- (a) Ciliated epithelium is a lining tissue found in the human gas exchange system. The tissue is composed of ciliated epithelial cells and goblet cells.

Squamous epithelium is a lining tissue found in the walls of the alveoli.

- (i) Squamous epithelial cells do **not** have cilia, unlike ciliated epithelial cells.

Suggest why ciliated epithelial cells are **not suitable** as the lining tissue in the walls of alveoli.

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- (ii) Goblet cells and mucous glands function to maintain the health of the gas exchange system. They produce mucus that traps pathogens and other particles, such as dust.

Explain why ciliated epithelial cells are also important in maintaining the health of the gas exchange system.

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- (b) The statements in Fig. 5.1 refer to the saturation of haemoglobin with oxygen in blood leaving the alveolar capillaries.

For a healthy person who inhales air in areas with **no air pollution**, the saturation of haemoglobin with oxygen can be as high as 99%.

For the same healthy person who inhales air in areas with **high air pollution**, the saturation of haemoglobin with oxygen is often lower than 99%.

Fig. 5.1

Suggest explanations for the differences stated in Fig. 5.1.

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